The Economics of Frontier-AI Competition: Leader Profitability under Open-Weight Catch-Up*

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Abstract

Frontier AI labs spend tens of billions of dollars a year training models whose capabilities an open-weight fringe reproduces within months and then sells at marginal compute cost. Because the leader can charge only for the capability increment it holds over that fringe, its ability to earn the model-building bill back rests on four rates and how they combine: how fast the frontier moves, how fast the follower closes the gap, how steeply value grows with capability, and how fast the bill grows. This note sets out a small dynamic model of that race, built in three cumulative levels: constant rates first; then a compute slowdown set against AI-accelerated algorithmic progress, and a bending value curve; then a follower with an engine of its own and two imitation channels. The model's work is to make the question precise rather than to settle it: under constant rates a single inequality between two growth rates decides the verdict for all time, and each mechanism added above that moves the date through a named dial. The contested parameters carry ranges rather than points, and the ranges move as the evidence does, so the numbers live where they can move: the companion interactive explorer holds the current calibration with its sources, the thresholds and dates it implies, and a Monte Carlo mode that turns the headline question into a forecast distribution. This note fixes the model those forecasts are read from.

1 The question

Frontier AI labs spend tens of billions of dollars a year training models whose capabilities an open-weight fringe reproduces within months and then sells at marginal compute cost. It is thus an open question whether the frontier labs can ever turn a profit: whatever they build is soon offered by others at a price that leaves no margin. This note sets out a small dynamic model of that race to answer it.

The model treats the frontier labs as a single leader and the open-weight ecosystem as a competitive fringe that prices at cost. The leader can then charge only for what it holds over the fringe: its earnings are the value of the frontier times the share of that value its lead protects — a share it holds on two counts, the capability the fringe cannot match at any price and the

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compute a distilled version of the leader's own model saves on matching what the fringe does have. Under what conditions does the leader's flow profit turn positive within roughly five years — and stay positive? The answer turns on four rates and how they combine: how fast the frontier moves, how fast the follower closes the gap, how steeply value grows with capability, and how fast the model-building bill grows.

The note mirrors the companion explorer exactly: Section 2 lays out the skeleton, and Sections 3–5 follow the explorer's three levels — Basics, Dynamics, Catch-up channels — switching mechanisms on cumulatively, so nothing is ever replaced, only added. Section 6 assembles the outputs, states the long-run benchmark, and says which dials carry the verdict.

2 Setup

Two players. The **leader** L is the set of frontier labs collapsed into one representative player training the single frontier model. The **follower** F is the open-weight / competitive fringe: it sells inference at marginal compute cost, so it earns nothing itself but disciplines the leader's price — customers can obtain the follower's capability essentially for free, and the leader can charge only for the increment above it.

States. Capability is measured in orders of magnitude (OOMs) of *effective compute*: algorithmic level times physical training compute. With $a = \log_{10}(\text{algorithmic level})$ and $c = \log_{10}(\text{training compute})$,

$$x_t^i = a_t^i + c_t^i \quad (i \in \{L, F\}), \qquad \Delta_t = x_t^L - x_t^F, \tag{1}$$

normalized so that the mid-2026 frontier sits at $x_0^L = 0$ — component-wise, $a_0^L = c_0^L = 0$; the follower starts a gap Δ_0 behind.

Payoff skeleton. Let $W(x)$ denote the *value index* of capability x: a dimensionless map with $W(0) = 1$, so W reads as value as a multiple of today's frontier value. The model carries no dollar level at all: earnings and the model-building bill are each normalized by their own level *today*, so every financial quantity is measured in multiples of today's model-building bill, and the leader's flow profit is simply

$$\Pi_t^L \;=\; E_t - B_t, \tag{2}$$

with earnings E_t the frontier's value times the share of it the lead protects, and the bill starting at $B_0 = 1$. The model's whole content is in the dynamics that generate x_t^L and x_t^F, the shape of W, the cost path, and the one financial parameter the normalization leaves behind.

3 Basics — the race, costs, and earnings

The base model holds every rate constant. The leader's capability grows at a constant speed: compute scales at rate g_c (OOM/yr) and algorithms improve at rate g_a, so $\dot{x}^L = g_c + g_a$. The follower has no engine of its own — it moves purely by catching up, closing the gap at rate δ (per year):

$$\dot{x}_t^F = \delta\,(x_t^L - x_t^F) = \delta\,\Delta_t, \qquad \text{so} \qquad \dot{\Delta}_t = \dot{x}^L - \delta\,\Delta_t. \tag{3}$$

The observed lag of the open-weight fringe behind the frontier pins δ: with $\delta\Delta_0$ equal to the leader's speed, the lag is constant and the gap holds at Δ_0.

Value is exponential in capability: each additional OOM multiplies the index by 10^{ν},

$$W(x) = 10^{\nu x}, \tag{4}$$

with the slope ν measured in OOMs of value per OOM of capability.

The bill. On the cost side, the leader rents compute and pays the bill B_t of the model it is currently running, in units of today's bill:

$$B_t \;=\; 10^{\,c_t^L - g_p t}, \tag{5}$$

where g_p is the hardware price-performance decline (OOM/yr) — the bill tracks the compute path at falling prices, and algorithmic progress is already counted inside x, not here. Since $c_0^L = 0$ by the capability normalization, $B_0 = 1$: the bill is measured in multiples of itself today.

Earnings. Earnings are the rent on the lead: the frontier's value — the market — times the share of it the lead protects, with the level pinned against today's bill,

$$E_t \;\sim\; \underbrace{W(x_t^L)}_{\text{market}}\;\underbrace{\big(1-\lambda^{-\Delta_t}\big)}_{\text{share protected}}, \qquad E_0 \;=\; \rho\, B_0, \tag{6}$$

where $\Delta_t = x_t^L - x_t^F$ is the lead in OOMs, $\lambda > 1$ is the factor by which the fringe's share of the frontier's value falls per OOM of lead (*Protection as a parameter*, below), and "earnings" means earnings *before* model-building costs. This is the earnings law at every level of the model, and it is the only one.

At this level it is the value gap itself. Value is exponential here, $W(x) = 10^{\nu x}$, and the derivation below sets $\lambda = 10^{\nu}$; then

$$W(x_t^L)\big(1-\lambda^{-\Delta_t}\big) \;=\; W(x_t^L) - W(x_t^L)\,10^{-\nu\Delta_t} \;=\; W(x_t^L) - W(x_t^F), \tag{7}$$

term for term. So the reader who prefers the difference may read (6) as (7) throughout Section 3 and lose nothing. The two part company only where the value curve stops being exponential (Section 4), and there it is (6) that the model runs.

The pair is exact as it stands: the $\sim$ hides one fixed constant, and the second equation pins it, so that equivalently $E_t = \rho\, B_0\, W(x_t^L)\big(1-\lambda^{-\Delta_t}\big)/\big(1-\lambda^{-\Delta_0}\big)$. With $B_0 = 1$ this makes $E_0 = \rho$, the model's one financial parameter, discussed next. Normalizing against today is how the level is pinned, here and everywhere below: every later way of writing the earnings line is the same path read as a ratio to its own value today, so the constants a $\sim$ carries never reach a result.

The gap is what a buyer gives up by defecting to the fringe, and it carries the leader's advantage on two counts rather than one. The first is *quality*: at the top of the ladder the fringe cannot deliver x^L at any price, so a buyer of frontier capability pays up to the value difference $W(x_t^L) - W(x_t^F)$ to keep it — vertical differentiation against a rival priced at marginal cost, which is the standard Bertrand outcome and, with tastes heterogeneous, still linear in the gap, the taste distribution entering only through a constant the $\sim$ absorbs. The second is *efficiency*: below the frontier the leader can serve capability the fringe already has, and serve it more cheaply. Capability is algorithms plus compute in OOMs, so the leader's algorithmic lead $a_t^L - a_t^F$ is literally a unit-cost ratio — a distilled, "mini" version of the frontier model reaches

a given capability on $10^{-(a_t^L - a_t^F)}$ of the compute the fringe needs for it — and undercutting a fringe that prices at its own cost leaves the leader a margin of $10^{\,y-a_t^F}\left(1-10^{-(a_t^L-a_t^F)}\right)$ per buyer of capability y. The two channels are not the same mechanism and do not share an elasticity; what they share is the object that governs them. Both are proportional to a value or cost level times a factor that is bounded in $[0,1)$, increasing in the gap, and zero when the gap closes, so both are carried by the lead and both vanish with it. Written as (6), the protected share is the efficiency channel's own margin fraction exactly — $1-10^{-(a_t^L-a_t^F)}$ is $1-\lambda^{-\Delta_t}$ with $\lambda = 10^s$, s the algorithmic share of the lead — and, by (7), the quality channel's exactly as well at $\lambda = 10^\nu$. One parameter therefore carries the mix of the two, and λ is where it is calibrated. At this level the choice between the channels' readings is empty, since both are exponential in the lead and any mix of them is one more constant the $\sim$ absorbs; above Basics it is not, and *Protection as a parameter* is where that is settled.

The coverage ratio. The headline object is the ratio of earnings to the bill,

$$\rho_t \;=\; \frac{E_t}{B_t}, \qquad \Pi_t^L > 0 \iff \rho_t > 1, \qquad \rho_0 = \rho. \tag{8}$$

The single financial parameter ρ is *coverage today*: operating earnings per unit of model-building outlay. Reported dollar figures are calibration evidence for it, not parameters of the model: with gross revenue R_0, operating margin before model-building costs m, and model-building outlay $k\,R_0$, coverage today is $\rho = m/k$ — at the filings the calibration currently reads, roughly a third: the industry earns some 33 cents per dollar of model-building spend. Dollar levels never enter: scaling every reported figure leaves the model unchanged, so only the ratio m/k was ever identified, and the model now carries just it. Break-even is $\rho_t = 1$; coverage starts at a handicap, and the question is whether it climbs.

Will the leader ever be profitable? With the gap constant and value exponential, one inequality decides it. Earnings inherit W's growth rate along the leader's path, growing by the factor $10^{\nu(g_c+g_a)}$ per year, while the bill grows by $10^{\,g_c-g_p}$ per year; coverage rises, and break-even eventually arrives, if and only if

$$\nu\,(g_c + g_a) \;>\; g_c - g_p. \tag{9}$$

Whether (9) holds is a calibration question, and a close one: both sides are growth rates of the same magnitude, each contested, and the companion explorer carries the current reading and the range around it. What the base model settles is the shape of the answer rather than its sign. Because every rate is constant, whichever way the inequality goes is one verdict for all time — and either way it rests on today's compute scaling persisting forever, exactly the assumption the Dynamics level drops. Dropping it does not settle the question either way: a compute slowdown bends the bill, but it also slows the frontier and lets the follower close the gap — and earnings live on the gap — while accelerating algorithmic progress pulls in the opposite direction. Whether break-even ever arrives becomes a quantitative race between these forces.

What the earnings equation assumes. The two factors of (6) carry quite different assumptions, and it is worth separating them. The second, $1-\lambda^{-\Delta_t}$, is the quality-ladder rent of the innovation literature (Grossman–Helpman; Aghion, Harris, Howitt and Vickers), bounded in $[0,1)$, increasing in the lead, and constant whenever the lag is constant — and it is

where both channels named above live, the quality premium at the frontier and the compute a distilled model saves below it, since each is governed by the same Δ_t and each vanishes with it. The first factor is the value of the frontier itself, and on a constant-lag path it does all the work: earnings grow only because the whole value curve climbs, never because the lead widens. The equation therefore asserts that the leader's market grows with the frontier's value — willingness to pay per customer rises with capability, while the customer base is held fixed — and the race inequality (9) is a statement about that market factor alone. That is a deliberate departure from the quality-ladder convention, in which the leader's rent is bounded and independent of the absolute technology level; the polar case, with the market factor held fixed, is value affine in capability, and there coverage falls at the bill's own rate $g_c - g_p$ for any value slope and any lag, so break-even never comes. Everything the model allows beyond that floor is bought by the convexity of W.

The general form of the assertion is elementary: along a path with constant lead Δ the protected share is constant, so $E_t \propto W(x_t^L)$ and earnings grow at $w'(x_t^L)\,\dot{x}_t^L$ in logs — accelerating, steady, or decaying according as $w = \log_{10} W$ is convex, affine, or concave in capability. Harvest versus commoditization is a statement about the shape of the value curve, and the object it turns on is the value slope w' that Section 4 bends. Above Basics the value curve is no longer exponential (Section 4), and there the two factors of (6) separate: the market factor $W(x_t^L)$ carries the bend, while the protected share does not, since it is a closed form in the lead and reads no value level at all. That separation is the substance of the next paragraph. The companion explorer states the law the same way at every level, and at its base level prints (7) beside it.

Protection as a parameter. At $\lambda = 10^{\nu}$ the protected share erodes with the lead exactly as fast as value grows with capability — one OOM of lag surrenders a factor 10^{ν} — and that is not a calibration but a consequence of the one case the quality channel treats: a buyer surrendering the entire value difference to a leader facing a fringe priced at marginal cost. Whatever makes real protection erode faster than the value difference alone — price competition among near-frontier substitutes — or slower — switching costs, integration, brand — is a departure from that case, as is any weight on the efficiency channel, whose own rate $\lambda = 10^{s}$ is set by the algorithmic share of the lead rather than by what capability is worth. The model absorbs all of it into the one parameter $\lambda > 1$ already carried in (6), and the reason the absorption is legitimate is that both channels are exponential in the lead: a mixture of $1 - \lambda_1^{-\Delta}$ and $1 - \lambda_2^{-\Delta}$ is of the same form to first order, with a λ between them. One departure of the slower kind is worth naming because it is an economics rather than a label: when serving compute is dear, a weaker model burns more of it to reach the same answer, so demand concentrates on the frontier and the lead protects more of its value than the quality difference alone would give it (Patel, 2026) — the Alchian–Allen mechanism, shifting the weight in the quality/efficiency mix toward efficiency as compute grows scarce. Whether it holds is an argument about the world and not a finding of this model; it is recorded here because λ is where such an argument would enter.

Reading the share on the lead rather than on the value ratio $W(x_t^F)/W(x_t^L)$ is a modelling choice, and above Basics it is a consequential one: the two coincide wherever W is exponential and part where it bends. The choice follows the economics. The margin the leader earns against a fringe priced at its own marginal cost is a *cost*-side object — the efficiency channel's $1 - 10^{-(a_t^L - a_t^F)}$ — and a cost-side margin does not shrink because the marginal buyer's willingness to pay for the

next OOM has begun to flatten. At a constant lead it is constant, which is what (6) says and what the value-ratio reading denies. Both limits of the parameter are finite and both mean something: as $\lambda \to 1$ the protected share becomes proportional to the lead measured in OOMs, and as $\lambda \to \infty$ protection never erodes at all and earnings are the market factor alone.

The parameter prices a gap that *moves*. On a constant-lag path the protected share is the same at every date, so it is one more constant the $\sim$ absorbs and λ leaves earnings exactly where they were — which is the Basics level, where the calibration holds the lag stationary by construction. It therefore does nothing at all in Section 3 and bites only once the gap itself is in motion, which is where the companion explorer first offers the choice.

Nothing in the evidence identifies λ, and the obstruction is structural rather than a gap to be filled. The fringe prices at marginal cost here, so the value it commands is surplus that is never billed, and the denominator of the protected share — the frontier's whole value — is not an accounting quantity either. Price premia and token-share splits between frontier and open models can be read as λ only through assumptions this model does not make, and read that way they disagree several-fold with each other and with themselves over time. The default therefore binds λ to 10^{ν}. The free parameter is carried so that the cost of the assumption is visible rather than silent: the explorer sweeps it across the widest range any reading in that evidence supports and reports its leverage beside the other dials'. Its direction is structural, not calibrated: a larger λ keeps more of today's protection while the fringe is closing the gap, and less of the gain while the lead is widening, so it helps wherever the lead narrows over the window and costs where it widens. The race inequality (9) is untouched either way.

4 Dynamics

The Dynamics level turns on two opposing forces — compute growth slows down while algorithmic progress speeds up — together with a bend in the value curve itself. The follower, earnings, and profit carry over from Basics unchanged.

The transition curve Γ

Three of the model's mechanisms are *transitions*: something moves from one steady value to another, over time or over capability. Two appear in this section, and a third with the follower's compute in the next; all use the same primitive, stated once: the logistic S-curve, dialed in four *observables*,

$$\Gamma\big(u;\, y(0),\, y_\infty,\, u_{mid},\, q_0\big)\,: \tag{10}$$

today's value $y(0)$, the destination y_∞, the midtime u_{mid} at which the transition is halfway done, and the position q_0 — the fraction of the transition already completed today (necessarily $q_0 < \frac{1}{2}$: the midpoint is still ahead). Figure 1 shows the shape. The plateau the curve came from and its steepness are *not* free parameters: they are derived so that the curve passes through today's value, already q_0 of the way along, at $u = 0$ (Appendix A has the closed form). So $\Gamma(0) = y(0)$ identically — an observation never depends on a scenario parameter, and moving u_{mid} or q_0 says where the transition *started*, never what it is now. By the logistic's symmetry the curve is $1 - q_0$ done at $2u_{mid}$, so a transition occupies roughly the window $[0,\, 2u_{mid}]$; and whenever $y_\infty = y(0)$ it is exactly constant.

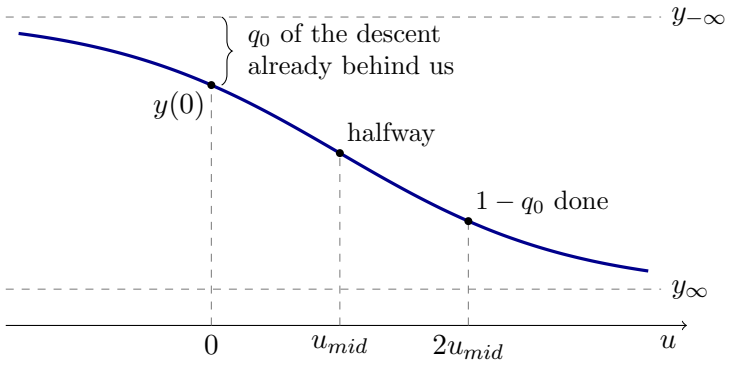


Figure 1: The transition curve $\Gamma(u;\, y(0),\, y_\infty,\, u_{mid},\, q_0)$, drawn for a falling transition (the compute slowdown's shape) with $q_0 = 0.25$. The four arguments are observables; the pre-transition plateau $y_{-\infty}$ and the steepness are derived (Appendix A), so the curve passes through today's value $y(0)$ at $u = 0$ for any choice of u_{mid} and q_0.

Force 1 — the compute slowdown

Today's compute scaling cannot persist: power, fabrication capacity, and capital all bind. Compute growth now rides the transition curve down to a long-run floor $g_{c\infty}$, half-done at t_{mid}:

$$\dot{c}_t^L \;=\; g_{c,t} \;=\; \Gamma\big(t;\, g_c,\, g_{c\infty},\, t_{mid},\, q_0^c\big). \tag{11}$$

The anchor is *today's* realised growth g_c — the same g_c as in Basics, held fixed by construction — so positing that the slowdown is further along (q_0^c larger) raises the pre-slowdown plateau it came down from, never today's growth. Whenever $g_{c\infty} = g_c$ the curve is exactly constant and the base model never sees the shape.

Force 2 — algorithmic progress speeds up

The base model's constant algorithmic rate g_a becomes a *produced* quantity, the product of three factors: a difficulty drag, researcher effort accelerated by AI assistance, and experiment throughput, which moves with compute growth:

$$\dot{a}_t^L \;=\; g_a\, 10^{-\beta_h (a_t^L - g_a t)} \left(\frac{\psi(x_t^L)}{\psi(0)}\right)^{\lambda_R} \left(\frac{g_{c,t}}{g_c}\right)^{\alpha_W}, \qquad \psi(x) = 1 + \beta_0\, 10^{\,\gamma x}. \tag{12}$$

Both ratios are normalized to one at $t = 0$ and the drag starts at one, so $\dot{a}_0^L = g_a$ holds identically: today's rate remains an observation, never a parameter. The function ψ is the AI-assistance multiplier: $\psi(0) = 1 + \beta_0$ is the counterfactual multiplier on frontier-lab research *throughput* today, so $\beta_0/(1+\beta_0)$ is the share of that throughput AI assistance is responsible for — an input level, not an output speedup, the two differing through the elasticity the input carries below. And γ is how strongly that assistance compounds with capability — $\gamma = 0$ freezes AI assistance at today's level. (Lower-case γ, a number, is unrelated to the transition curve Γ.)

The three exponents are each a published elasticity, not a fitted share. The drag $10^{-\beta_h (a_t^L - g_a t)}$ is "ideas getting harder to find": β_h — the subscript is for *harder*, keeping it apart from the AI share β_0 in the same display — measures how research productivity falls as the algorithmic

level runs ahead of the balanced path $a = g_a t$. Because the drag is measured against that path rather than against the level, it bites only off balance: at $\gamma = 0$ and constant compute growth the balanced path $a = g_a t$ solves (12) exactly, for every β_h. λ_R is the elasticity of the algorithmic rate to the research input ψ — the subscript marks it as a *research* elasticity, a different object entirely from the protection parameter $\lambda = 10^{\nu}$ of Section 3. And α_W is the elasticity of the algorithmic rate to experiment-compute *growth*: the compute channel enters as a growth rate, not a stock, so a slowdown in $g_{c,t}$ reaches the algorithmic engine directly.

With $\gamma > 0$ the ψ-feedback compounds: capability raises the research input, which raises the algorithmic rate, which raises capability. Whether the loop runs away is a parameter condition, not a statement about any horizon. Writing $\lambda_R = r_R \beta_h$ — with r_R the ratio of the research elasticity to the difficulty exponent, the returns-to-research ratio of the growth literature — the path is explosive, reaching a finite-time singularity, if and only if

$$\gamma \lambda_R \;>\; \beta_h \qquad \Longleftrightarrow \qquad \gamma r_R \;>\; 1, \tag{13}$$

and sub-critically it settles instead to the quasi-balanced rate

$$g_\infty \;=\; \frac{\beta_h g_a + \gamma \lambda_R g_{c\infty}}{\beta_h - \gamma \lambda_R} \;=\; \frac{g_a + \kappa g_{c\infty}}{1 - \kappa}, \qquad \kappa = \gamma r_R, \tag{14}$$

once compute growth has reached its floor. The second form shows β_h dividing out: the difficulty exponent sets how fast the path reaches the quasi-balanced regime, never where it lands. The calibration the explorer ships is sub-critical, and the explorer flags when the feedback's share of algorithmic progress stops being small. More consequential for the race, the factor $(g_{c,t}/g_c)^{\alpha_W}$ couples algorithmic progress to compute growth, so the slowdown drags the algorithmic engine down with it rather than leaving it untouched.

Equation (12) is the standard law of the quantitative AI-progress literature, not a bespoke one: it is the semi-endogenous kernel of Jones (1995), as measured by Bloom et al. (2020) — β_h is numerically their difficulty exponent — read in the model's log units, with the compute channel entering as a growth rate, the reduced form Whitfill et al. (2025) validate, and with ψ playing the role of automation-augmented research labor. The knife-edge (13) is the condition the recursive-self-improvement literature states in several coordinate systems; Appendix B gives the mapping term by term.

Value in logs — the slope transition

Write value in logs, $w = \log_{10} W$, so $W(x) = 10^{w(x)}$ with $w(0) = 0$ (the index anchor $W(0) = 1$). What the Dynamics add is that the *slope* of w rides the transition curve — its second use, this time over capability rather than time:

$$w'(x) \;=\; \Gamma\big(x;\, \nu,\, \nu_\infty,\, x_{mid},\, q_0^w\big). \tag{15}$$

Each OOM of capability is worth $10^{\nu}\times$ *today* ($w'(0) = \nu$, the same ν as in Basics), easing to a floor of $10^{\nu_\infty}\times$ once the transition is done, halfway at x_{mid} (OOMs above today's frontier). There is no hard ceiling: value grows without bound at the floor slope, so each further OOM is worth at least $10^{\nu_\infty}\times$.

The shape of the value curve is the model's pivotal assumption. On the steep stretch a defended, constant gap earns *growing* rents — the leader harvests its lead. Once the slope eases, the

same gap's rents grow ever more slowly: commoditization in all but name, if ν_∞ is low. Where the transition sits (x_{mid}) and where the slope lands (ν_∞) are therefore decisive *as a pair*, and asymmetrically so: x_{mid} is the one that gates the other. If the midpoint sits far above the capability the frontier reaches within the horizon, the whole path runs on the steep stretch and the floor slope barely registers, however low it is set; bring the midpoint within reach and the same choice of ν_∞ swings the outcome between harvest and commoditization. So the two are explored jointly, with x_{mid} deciding whether ν_∞ matters at all. The asymmetry has to be stated carefully, because it says which *question* each dial answers and not which way the verdict goes. Between them the pair decides how much of the horizon is spent on the steep stretch, and so how much the floor slope is allowed to matter at all — and moved across the range the calibration allows, either dial can decide whether the leader's flow profit turns positive inside the horizon at all. Where the gate sits at the current calibration, and how close the shipped midpoint runs to it, is the explorer's to report. What decides the sign outright under constant rates is still the *level* of the value-growth rate ν.

The frontier speed race

The frontier's speed is the sum of the two forces,

$$\dot{x}_t^L \;=\; \dot{c}_t^L + \dot{a}_t^L, \tag{16}$$

and they pull in opposite directions: the compute slowdown brakes, the ψ-feedback accelerates — and through the experiment factor $(g_{c,t}/g_c)^{\alpha_W}$ the brake also drags on the accelerator. Whether the frontier speeds up or slows down is a calibration question, not a modeling one. One consequence carries forward: with δ still pinned by the observed lag, a decelerating leader means a *closing* gap. The next level gives the follower an engine of its own, which re-anchors the gap on the observed lag *today* — but an engine of its own is no guarantee that the gap stays open, and whether it closes over the horizon becomes a quantitative question in its turn.

5 Catch-up channels

The follower now gets what the leader has had from the start: its own algorithmic rate g_a^F and its own compute path, the same transition-curve family with its own parameters,

$$\dot{c}_t^F \;=\; \Gamma\big(t;\, g_c^F,\, g_{c\infty}^F,\, t_{mid}^F,\, q_0^F\big), \tag{17}$$

and the single catch-up rate δ unpacks into two channels. What catches up is the follower's *algorithmic* level, because compute cannot be copied:

$$\dot{a}_t^F \;=\; g_a^F \;+\; \underbrace{\delta_a\left(a_t^L - a_t^F\right)}_{\text{published methods}} \;+\; \underbrace{\delta_x\left(x_t^L - x_t^F\right)}_{\text{distillation}}. \tag{18}$$

Catch-up through *published methods*, at rate δ_a, is ambient diffusion — papers, espionage, talent mobility, know-how — so it runs on the algorithmic gap and operates even if the leader releases nothing. *Distillation*, at rate δ_x, copies what the leader's deployed model can do, so it runs on the capability gap. Both gaps are assumed positive in (18) — the leader is ahead; in the implementation each gap is floored at zero, so imitation dies at parity and the follower

never un-learns. Both cross terms measure imitation only; self-improvement gains run on each player's own engine.

The initial gap splits between the two states,

$$a_0^F = -\,\text{split}\cdot\Delta_0, \qquad c_0^F = -(1-\text{split})\,\Delta_0. \tag{19}$$

The calibration treats the channel rates and the initial split differently. The observed open–closed lag pins the $t=0$ *transfer*: with the follower supplying most of its own speed, the two channels together need only close the wedge between the leader's speed and the follower's own engine, $\dot{x}_0^L - \left(g_a^F + \dot{c}_0^F\right)$, spread over the gap Δ_0. Because catch-up through published methods runs on the algorithmic gap, which at $t=0$ is $\text{split}\cdot\Delta_0$ rather than Δ_0, what the lag pins is the *weighted* sum $\text{split}\cdot\delta_a + \delta_x$, not the bare total. What the lag leaves free is the *direction*, and that direction is itself an observable: let

$$s \;\equiv\; \frac{\delta_x\,\Delta_0}{\dot{x}_0^L - \left(g_a^F + \dot{c}_0^F\right)} \tag{20}$$

be the share of the transfer that runs through the models the leader *serves* — operationally, what would stop if the leader trained the model but never served it. Then both coefficients are written by dividing the transfer and converting each half into a rate on the gap it reads,

$$\delta_x = \frac{s\left[\dot{x}_0^L - (g_a^F + \dot{c}_0^F)\right]}{\Delta_0}, \qquad \delta_a = \frac{(1-s)\left[\dot{x}_0^L - (g_a^F + \dot{c}_0^F)\right]}{\text{split}\cdot\Delta_0}, \tag{21}$$

and the stationarity condition holds identically in s. The Level-3 follower is therefore identified by three observables and no free coefficients: the lag fixes the gap and the transfer, the follower's own-research share removes its engine from that transfer, and s divides what remains. Neither δ_a nor δ_x is set directly. The initial-gap split remains a free, explored parameter, but it is now a bookkeeping one: at held s it converts the methods half of the transfer into a coefficient and drops out of the flows themselves.

Only s carries the release-delay question, since withholding acts on the served channel alone; the explorer states its evidence and the range the forecasts are swept over.

6 Outputs

The model is now complete: the leader's path from (11)–(12), the follower's from (17)–(18), value from (15), and profit from the skeleton (2) with cost (5).

Headline. The headline output is the path of the leader's flow profit Π_t^L: does it turn positive within roughly five years, and does it *stay* positive? Equivalently, does the coverage ratio $\rho_t = E_t/B_t$ cross one — starting from its handicap $\rho_0 = \rho$? Every verdict the model produces is about that flow and nothing else: the yearly margin, whether its sign turns, and the date at which it turns. "Profitable" means that sign, at that date.

The long-run benchmark. When the system settles — bounded growth regime, constant gap — earnings grow at $\nu_\infty\,\dot{x}_\infty^L$ and the bill at $g_{c\infty} - g_p$ (both in OOM/yr), so in the long run the leader harvests if and only if

$$\nu_\infty\,\dot{x}_\infty^L \;>\; g_{c\infty} - g_p. \tag{22}$$

The normalizations are constant factors and drop out of growth rates. This limit favors harvest at a strikingly low floor slope ν_∞: once compute growth has fallen to its floor, even modest value scaling outruns the bill. The battleground is therefore the *transition* — today the bill's growth outruns earnings growth for moderate ν, so Π_t^L dips before recovering, and the five-year sign question is about levels and the slowdown timing t_{mid}, not the limit.

What decides it. The model's work is to say which dials the verdict turns on, and through what channel each acts. Under constant rates the sign is settled outright by the race inequality (9), and the dial with the most leverage on it is the value-growth rate ν: both sides of the inequality are growth rates of the same magnitude, so a modest revision in what a year of capability is worth can turn it. Above Basics, mechanism joins the rates as a mover of the date: the compute slowdown bends the bill and drags the algorithmic engine down with it (12); the value curve's midpoint x_{mid} and floor ν_∞ decide as a pair how much of the horizon is spent on the steep stretch (Section 4); the follower's own engine and the two imitation channels govern the gap that earnings live on. On the financial side, coverage today ρ sets the handicap the race starts from, and the hardware price-performance decline g_p works on the bill alone, so (9) can turn with no change at all in what capability is worth. Each of these is an observable the calibration tracks, so a verdict read off the model is falsifiable dial by dial.

Where the numbers live. This note deliberately states no verdict. The dials above are contested — each carries a range rather than a point, and the ranges move as the evidence does — and at readings the current evidence supports, the two sides of (9) run close. A single path is therefore the wrong object to argue over: the honest answer to the headline question is the distribution drawn when every contested range is sampled at once, and the sign of the verdict should be read off that fan rather than off any one line. That is the companion explorer's[1] job: it implements this model level by level — Basics, Dynamics, Catch-up channels — documents every calibration target and its sources, flags every parameter above as calibrated to a stated observable or free, reports the thresholds and break-even dates the current calibration implies, and draws jointly from all ranges in a Monte Carlo mode to turn the headline question into a forecast distribution. The verdicts live there, at the calibration of the day; what this note fixes is the model they are read from.

Appendix A The transition curve in closed form

Let σ be the sigmoid function, $\sigma(z) = 1/(1+e^{-z})$, rising from 0 to 1 with $\sigma(0) = \frac{1}{2}$, $\sigma(-z) = 1 - \sigma(z)$, and inverse the log-odds $\sigma^{-1}(f) = \ln \frac{f}{1-f}$, read on a fraction f. The transition curve is the sigmoid run between two plateaus,

$$\Gamma\big(u;\, y(0),\, y_\infty,\, u_{mid},\, q_0\big) \;=\; y_{-\infty} + \big(y_\infty - y_{-\infty}\big)\,\sigma\big(k\,(u - u_{mid})\big), \tag{23}$$

with the steepness k and the pre-transition plateau $y_{-\infty}$ derived from the four dialed observables by two identities.

The steepness, from the position. The fraction of the transition completed at $u = 0$ is $\sigma(-k\,u_{mid}) = q_0$, so

$$k\,u_{mid} \;=\; \ln \frac{1-q_0}{q_0}, \tag{24}$$

[1] https://pkocourek.com/frontier_labs_profit

the log-odds of the journey remaining. The curve then reads q_0 done at $u = 0$, one half at u_{mid}, and — by the sigmoid's symmetry — $1 - q_0$ at $2u_{mid}$, for any q_0.

The plateau, from today's value. Today's value is the position-weighted average of the two plateaus, $y(0) = (1 - q_0)\, y_{-\infty} + q_0\, y_{\infty}$; un-mixing it gives

$$y_{-\infty} \;=\; \frac{y(0) - q_0\, y_{\infty}}{1 - q_0}. \tag{25}$$

As $q_0 \to 0$ the plateau is today's value — nothing has happened yet. For a slowdown ($y_{\infty} < y(0)$) it returns $y_{-\infty} > y(0)$, correctly: we have already come part of the way down.

Substituting both identities into (23) confirms $\Gamma(0) = y(0)$ for every u_{mid} and q_0.

Appendix B Relation to standard forms

The algorithmic law (12) is assembled from published forms, and this appendix states the assembly precisely. The symbols λ, β, ϕ, and R are the literature's own and are local to this appendix; in particular this λ is unrelated to the protection parameter $\lambda = 10^{\nu}$ of Section 3.

The kernel. The quantitative literature on research-driven progress runs on one law, the semi-endogenous kernel

$$\dot{A}_t \;\propto\; R_t^{\lambda}\, A_t^{\phi}, \qquad \beta \;\equiv\; 1 - \phi, \tag{26}$$

with A the technology level, R a research input, λ the returns to research effort, and β the rate at which ideas get harder to find (Jones, 1995; Bloom et al., 2020); Aghion et al. (2019) automate the research input, and Jones (2026) surveys the AI reading. In the model's units $A = 10^{a}$, so $10^{-\beta_h a} = A^{-\beta_h}$: the drag in (12) is the kernel's difficulty term read in logs, and β_h is *numerically identical* to the β that Bloom et al. (2020) publish — their estimates are priors for it, with no rescaling. That identity is the calibration payoff of writing the law this way. The one liberty (12) takes is to measure the drag against the balanced path $a = g_a t$ rather than against the level: the input growth that historically offset rising difficulty is bundled into g_a, so the calibrated baseline is a quasi-balanced path of the kernel and the drag prices only deviations from it.

The compute channel. The kernel feeds R a stock; (12) feeds the experiment channel a growth *rate*, $(g_{c,t}/g_c)^{\alpha_W}$. That is the reduced form of Whitfill et al. (2025), whose Proposition 1 — capability growth proportional to research-compute growth, with elasticity α_W (the subscript credits them) — is the kernel's own limit at returns ratio $\lambda/\beta = 1$, where growth in maps to growth out. What the reduced form buys is exactly what this model uses: the closed form of the compute path and the $t = 0$ normalization $\dot{a}_0^L = g_a$. What it costs is that $\lambda/\beta = 1$ is hard-wired into that channel: a compute slowdown understates the hit to $\dot{a}^L$ if the true ratio exceeds one, and overstates it if the ratio falls short.

The research input. $\psi(x) = 1 + \beta_0\, 10^{\gamma x}$ is automation-augmented research labor: human researchers (the 1) plus AI assistance whose supply scales with capability. β_0 is the automated share of research throughput today — the counterpart of the automated share f in Davidson et al. (2026) — and γ is the capability-elasticity of research input, the loop elasticity of Cunningham et al. (2026); because its historical identification is weak (Trammell, 2026), the model treats it as a dial rather than an estimate.

The exponents and the knife-edge. $\lambda_R = r_R\,\beta_h$, where r_R is the returns-to-software-R&D ratio λ/β that Besiroglu et al. (2024) estimate, so the criticality condition (13) is free of β_h: explosive if and only if $\gamma\,r_R > 1$. That is the literature's own knife-edge, stated in three coordinate systems. Whitfill and Wu (2025) state it as $\phi + \lambda > 1$ when research inputs substitute: their research input grows one-for-one with A, so the condition is $\lambda > \beta$ — the case $\gamma\,r_R > 1$ at unit loop elasticity. Davidson et al. (2026) state it as $\sum_i f_i\,r_i > 1$, automated shares weighting returns ratios across feedback channels, summed against one — the accounting descends from the compute-centric framework of Davidson (2023). And Cunningham et al. (2026) state it as a product of elasticities around the loop exceeding one — research input gained per unit of capability, times algorithmic rate gained per unit of research input, discounted by the difficulty term — which is $\gamma\,\lambda_R/\beta_h$ read one factor at a time. The same knife-edge, in shares, in ratios, or in elasticities.

What the model deliberately does not do. Compute is exogenous — the endogenized benchmark, with investment responding to AI revenue, is Erdil et al. (2025), cited rather than built — and ψ's interior carries no automation ceiling: nothing caps the share of research AI can do, where Davidson et al. (2026) bound it explicitly.

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